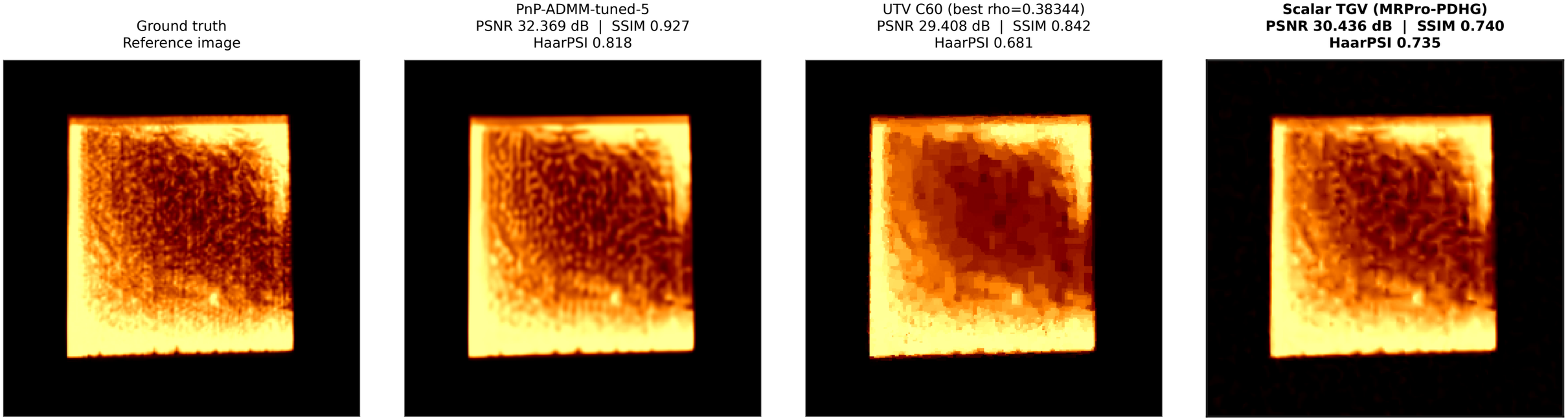
Perovskite 0292 | Scalar TGV compared with PnP-ADMM and optimised C60

One frozen 20% Hadamard acquisition | NPL-default PCM noise | identical GT

TGV parameters selected using this image's ground truth PSNR (single-image oracle)



TGV: alpha1 = 0.02828427, alpha0 = 0.01, alpha0/alpha1 = 0.3536, 8,192 PDHG iterations

UTV C60: global rho = 0.3834424; C60 original map, 512 PDHG iterations. PnP-ADMM: fixed tuned-5 parameters.

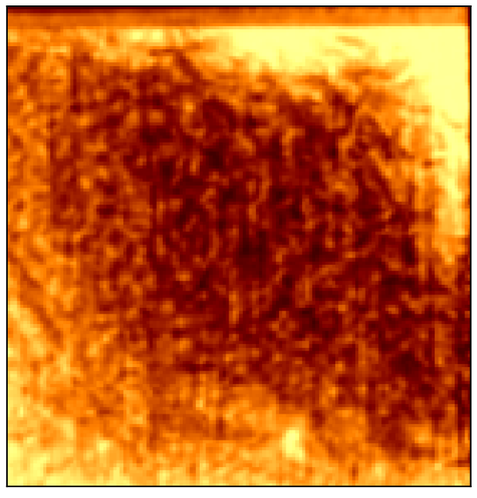
TGV uses the existing LION AdaptiveTGVPCMNetwork with MRPro-based PDHG and componentwise L1 TGV terms; the two weights were tuned independently.

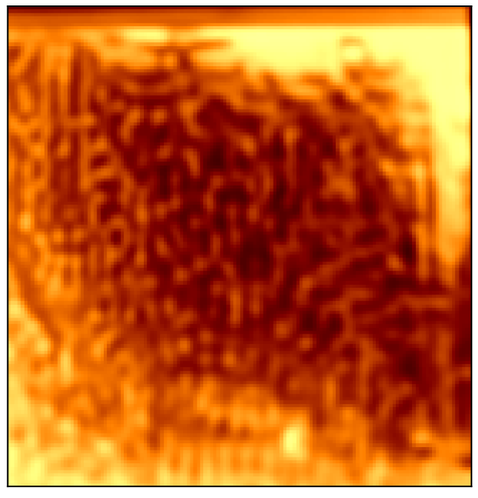


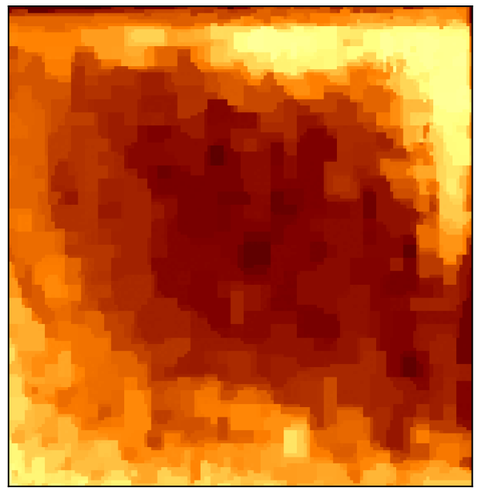
Shared display [0,1]; metrics on unclipped images

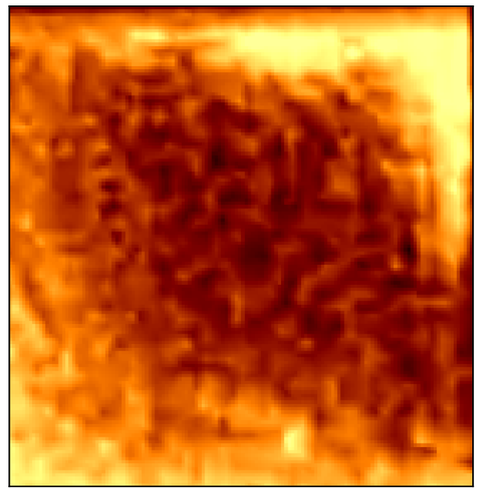
Perovskite 0292 | Zoomed texture and TGV parameter selection

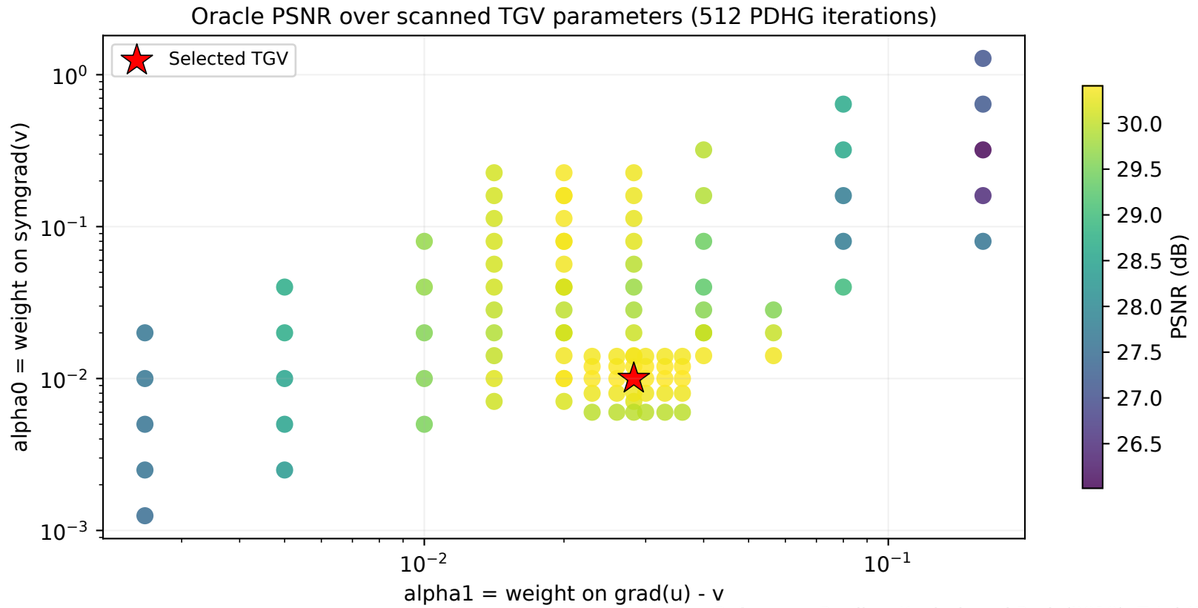
Crop: rows 40:185, columns 65:205; same region and intensity scale in every method











TGV parameter search and convergence
Coarse: 35 pairs at 512 iterations
Refined grid: 39 pairs at 512 iterations
Local refinement: 30 pairs at 512 iterations
Validation: strongest candidates at 2,048 / 4,096 / 8,192 iterations

Selected alpha1 = 0.0282842712
Selected alpha0 = 0.01
PSNR = 30.4363 dB
SSIM = 0.7402
HaarPSI = 0.7345

Difference vs UTV C60: +1.0280 dB
Difference vs PnP-ADMM: -1.9331 dB

Oracle tuning: GT is used to choose alpha1 and alpha0.
Comparison across methods has different iteration costs.

Reference: Bredies, Kunisch and Pock (2010), Total Generalized Variation, DOI 10.1137/090769521. MRPro: docs.mrpro.rocks

The tuned TGV penalty is componentwise L1 on gradient and symmetrised gradient; it is not a globally optimal TGV setting.